Area of $(x-3)^2+(y+2)^2<25:(x,y)\in L_1\cap L_2$

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The bounty expires in 6 days. Answers to this question are eligible for a +100 reputation bounty. Sid wants to **draw more attention** to this question:

> The existing answer involves envelopes, but the answer I am looking for should be along the lines of Tavish's comments. Equate the lines and find locus of intersection point. The bounds of the coordinates can be separately found...

Two lines (L_1, L_2) intersects the circle $(x-3)^2+(y+2)^2=25$ at the points (P, Q) and (R, S) respectively. The midpoint of the line segment PQ has x-coordinate $-\frac{3}{5}$, and the midpoint of the line segment RS has y-coordinate $-\frac{3}{5}$.

If A is the locus of the intersections of the line segments PQ and RS, then the area of the region A is:

What I've done:

Consider $L_1: y=ax+b$. The midpoint of the chord PQ is $(-\frac{3}{5}, -\frac{3a}{5}+b)$. Now, using the property that the midpoint of a chord of a circle and the center of the circle $(3,-2)$ are perpendicular we have:

$$\frac{-\frac{3a}{5}+b-(-2)}{-\frac{3}{5}-(3)}*a=-1$$

$$\implies b=\frac{3a^2-10a+18}{5a}$$

This means we can eliminate one variable and write the equation of $L_1: y=ax+\frac{3a^2-10a+18}{5a}$

From this form of L_1 we can get the value of the minimum value of the y-intercept by differentiating b. Let $b=f(a)=\frac{3a^2-10a+18}{5a}$, $f'(a)=\frac{3a^2-18}{5a^2}$, $f'(a)=0 \implies a=\pm\sqrt{6}$. I just found this hoping we will get bounds on the y-intercept of L_1.

Now lets do the same process for L_2.

Consider $L_2: y=cx+d$. The midpoint of the chord RS is $(\frac{-\frac{3}{5}-d}{c}, -\frac{3}{5})$. Now, using the property that the midpoint of a chord of a circle and the center of the circle $(3,-2)$ are perpendicular we have:

$$\frac{-\frac{3}{5}+2}{\frac{-\frac{3}{5}-d}{5}-3}*c=-1$$

c

$$\implies d = \frac{7c^2 - 15c - 3}{5}$$

This means we can eliminate one variable and write the equation of $L_2 : y = cx + \dfrac{7c^2 - 15c - 3}{5}$

From this form of L_2 we can get the value of the minimum value of the y-intercept by differentiating d. Let $d = f(c) = \dfrac{7c^2 - 15c - 3}{5}$, $f'(c) = \dfrac{14c - 15}{5}$, $f'(c) = 0 \implies c = \dfrac{15}{14}$. I just found this hoping we will get bounds on the y-intercept of L_2.

Along with all this, we can set bounds when the line segment is just about to leave the circle (tangent to the circle).

What I can visualize:

Let X = union of all line segments PQ. Let Y = union of all line segments RS. Every point in the intersection of X and Y is a candidate intersection point of the lines L_1 and L_2. So A = $X \cap Y$

Edit 1: I saw the equation of L_1 varying as a varies on DESMOS and think the boundry of the union of all line segments PQ might be an outer circle.

calculus geometry algebra-precalculus circles

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edited Mar 31 at 13:30

asked Mar 31 at 11:17

Sid
925 5 16

I would just write the intersection point in (x, y) in terms of (a, c) and then try to eliminate the parameters to obtain the locus. You also need $(x-3)^2 + (y+2)^2 \leq 25$. – Tavish Mar 31 at 12:13

I tried that. U can an equation interms of a,c... Those techiques would work if the locus was a curve... Here clearly the locus is a region cuz we ultimately need to find a shaded area... So I think it's better if find all the line segments PQ and take their union. Same for RS. Intersection of this is the required region – Sid Mar 31 at 12:16

In the end, you should get an equation with x, y and either one of a or c. Then after some examination of the family of curves, it should *probably* be possible to reduce the problem to finding the area between the curves for which the free parameter takes its highest and lowest values. – Tavish Mar 31 at 12:20

lets say i find the intersection in terms of a and i get $x = f(a)$ and $y = g(a)$. After this I'll have to dump these values into the circle: $(f(a) - 3)^2 + (g(a) + 2)^2 < 25$ as the segments must intersect inside the circle... but here i don't think this will boil down to a quadratic inequality, so i think its better if we use the other approach. if its promising, please demonstrate your solution. – Sid Mar 31 at 13:02

Actually I don’t want to, as it involves too much algebra to be promising. – Tavish Mar 31 at 13:13

1 Answer

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What you need is the envelope γ_1 formed by lines L_1 and the envelope γ_2 formed by lines L_2.

As the equation of L_1 is $y = \left(x + \frac{3}{5}\right) a - 2 + \frac{18}{5a}$, differentiating w.r.t. a we get: $x + \frac{3}{5} - \frac{18}{5a^2} = 0$, which can be solved for a:

$$a^2 = \frac{18}{5x+3}.$$

Inserting this into the equation of L_1 we get (after some algebra):

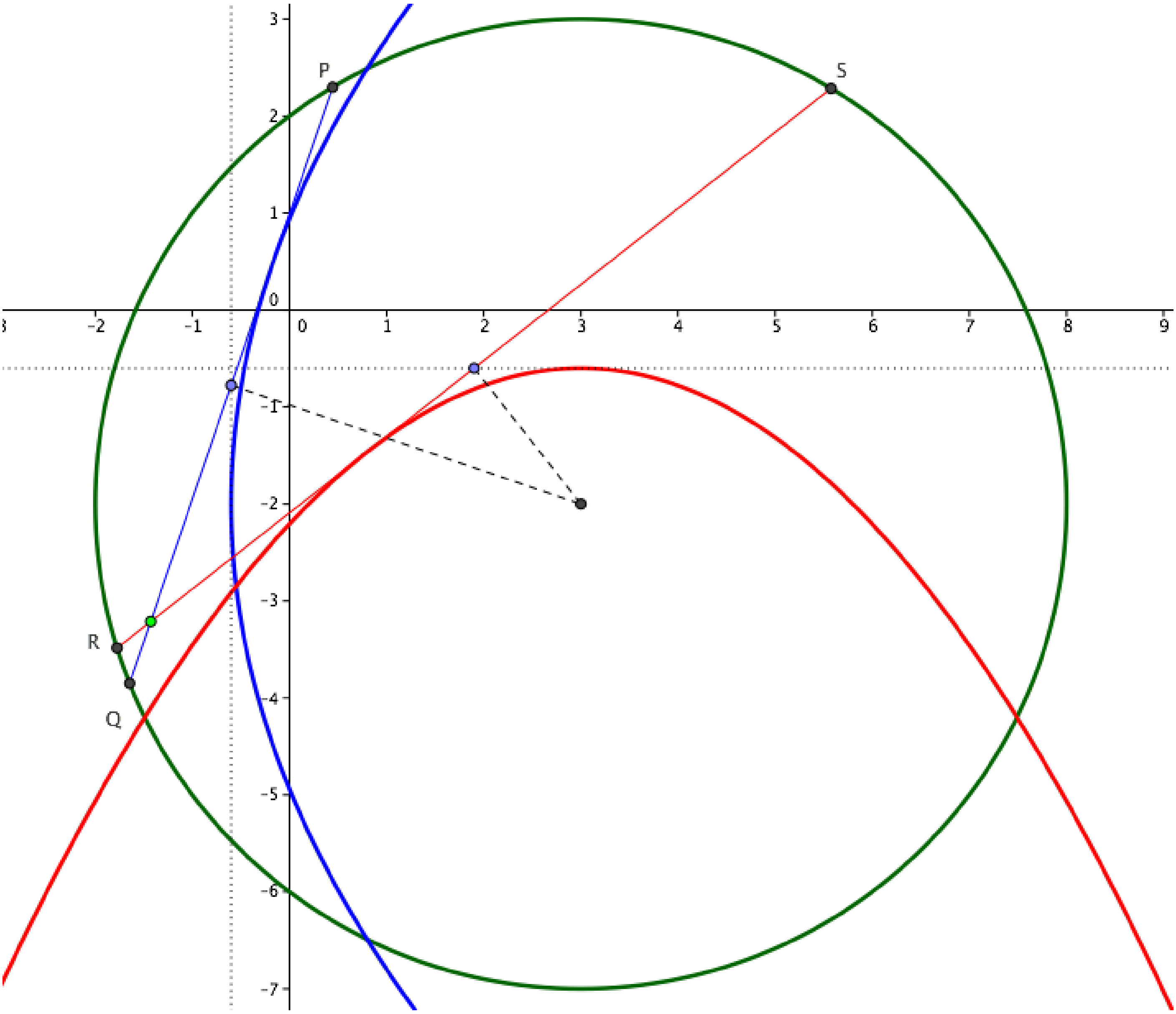
$$25(2+y)^2 = 72(5x+3),$$

which is the desired envelope γ_1 (a parabola).

Repeating the same process for L_2 we can find the equation of γ_2 (another parabola):

$$y = -\frac{5}{28}(3-x)^2 - \frac{3}{5}.$$

Lines L_1 and L_2 are tangent to their envelope, hence the area you want to compute is that external to both parabolas but inside the circle.



EDIT.

I want to show how this result can be obtained without calculus (see figure below).

Let's start with a chord AB of a circle of centre O. For any point M on that chord, we can construct a line RS perpendicular to OM. We want to find the envelope of all those lines, i.e. the curve which is tangent to all the lines.

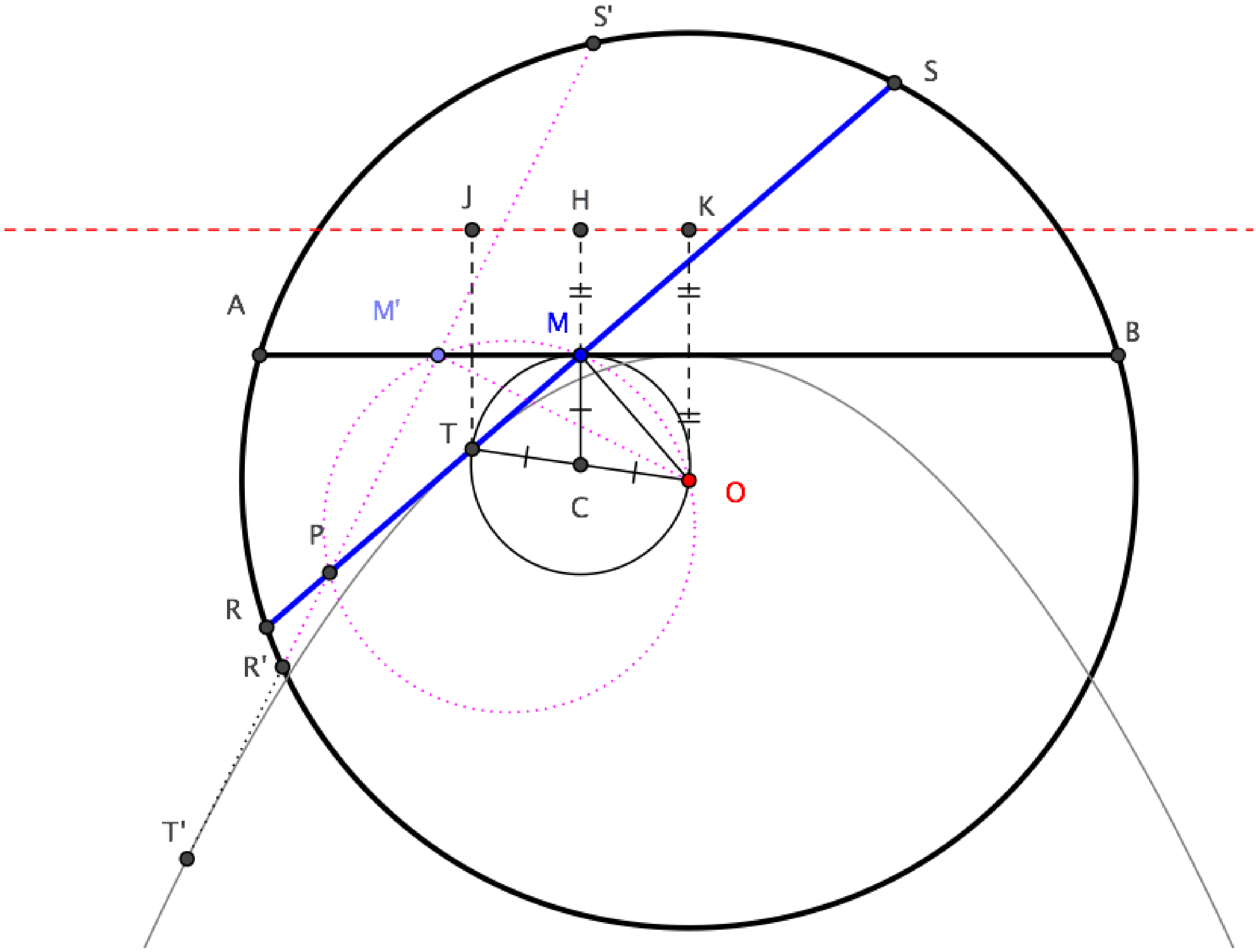
Consider then another point M' on AB and its associated line $R'S'$. Let P be the intersection of RS and

Consider then another point M' on AB and its associated line $R'S'$. Let P be the intersection of RS and $R'S'$, T the tangency point of RS with the envelope, T' the tangency point of $R'S'$ with the envelope. As M' approaches M, both T' and P approach T.

But the circle through OMM' also passes through P (because $\angle PMO = \angle PM'O = 90^\circ$) and this circle, as $M' \to M$, tends to the circle through O tangent to AB at M. Hence T, which is the limiting position of P, is the intersection of that circle with line RS. Moreover, OT is a diameter of that circle.

If we now construct line HK, parallel to AB at a distance from it equal to the distance of O from AB, and if J is the projection of T on it, then $TJ = TO$, because line CH joining the midpoints of the legs of a trapezoid is the arithmetic mean of bases OK and TJ.

It follows that point T has the same distance from O and from line HK, hence its locus is a parabola having O as focus and HK as directrix.



edited 5 hours ago

answered Mar 31 at 13:40

Intelligenti pauca

It's 18/5a right? You've written 18/a – Sid Mar 31 at 13:44

Yes, of course it's a typo, thank you: I'll correct at once. – Intelligenti pauca Mar 31 at 13:46

Isn't it a^2=18/(5x+3)? Not x+3? – Sid Mar 31 at 13:59

You are right, sorry, just corrected. – Intelligenti pauca Mar 31 at 14:02

yeah this seems to be right but i can seem to wrap my head around the reason why we can obtaining the envolope by diffrentiating wrt to a – Sid Mar 31 at 14:14